



MOHAMED ALSHAIKH

Mechatronics PhD Student · Deep Learning & Computer Vision · Applied AI & Intelligent Systems

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Arabic (Native) · English (C1) · Turkish (C1)

PROFESSIONAL SUMMARY

Mechatronics PhD student focused on **deep learning and computer vision**, with an engineering foundation in robotics, sensing, embedded platforms, and physical-system design. Experienced in developing and evaluating deep learning models for image classification, conducting experimental research, analyzing datasets, and communicating technical concepts. My academic background combines deep learning research with hands-on engineering projects involving computer vision, sensors, Arduino, Raspberry Pi, and mechanical design. Interested in **Computer Vision, Deep Learning, Machine Learning, and Applied AI & Robotics R&D**, particularly applications where intelligent computation interacts with real-world engineering systems.

EDUCATION

Ph.D. in Mechatronics Engineering

Sep 2025 – Present

Erciyes University · Kayseri, Türkiye

Research Focus: Machine Learning & Deep Learning Foundations, Computer Vision

- Developing deeper foundations in machine learning and deep learning, including their mathematical and theoretical foundations.
- Studying deep learning methods and their application to visual data.
- Conducting academic research, literature review, experimentation, and technical documentation.

M.Sc. in Mechatronics Engineering

Oct 2021 – Jul 2024

Erciyes University · Kayseri, Türkiye

Thesis: Optimizing Deep Learning Model Performance for Wireless Capsule Endoscopy Image Classification

- Developed and evaluated deep learning models for image classification using EfficientNet.
- Investigated data augmentation, model optimization, ensemble methods, and hyperparameter tuning.
- Achieved **95.75% classification accuracy** using ensemble methods and hyperparameter tuning.
- Conducted literature review, experimentation, model evaluation, dataset analysis, and scientific writing.
- Published research on data augmentation schemes for wireless capsule image classification based on EfficientNet.

B.Sc. in Mechatronics Engineering

Sep 2016 – Aug 2020

Düzce University · Düzce, Türkiye

GPA: 3.17 / 4.00 · Top 10 among department candidates

Graduation Project — Smart Garbage Separator Bin

- Developed a smart waste-sorting system for glass, metal, paper, and plastic.
- Integrated **Arduino UNO and Raspberry Pi** with capacitive, load-cell, and ultrasonic sensors.
- Implemented **OpenCV-based image processing** for non-metal waste separation.
- Designed mechanical components using **SolidWorks**.

TECHNICAL SKILLS

AI & Machine Learning

Deep Learning · Machine Learning · Computer Vision · Image Classification · Data Analysis

Programming

Python · C++ · C · MATLAB

Computer Vision & Engineering

OpenCV · Arduino · Raspberry Pi · SolidWorks · 2D/3D Modeling

Research

Literature Review · Experimental Research · Model Evaluation · Dataset Analysis · Scientific Writing · Technical Documentation

Professional

Project Coordination · Stakeholder Communication · Technical Communication

RESEARCH & PUBLICATIONS

- Published research on **deep learning and medical image classification** using EfficientNet.
- Experience with literature review, experimental design, model evaluation, scientific writing, and research documentation.
- Co-author of an additional research manuscript currently under review.

PROFESSIONAL EXPERIENCE

Coordinator & Export Specialist

May 2021 – May 2025

Jazeera Company · Gaziantep, Türkiye

- Coordinated projects involving clients, suppliers, and project partners across Türkiye and the Middle East.
- Managed project communication, documentation, progress reporting, and coordination.
- Supported technical equipment, infrastructure, and logistics projects.
- Conducted market research and supported business development activities.

Robotics Trainer — Part-Time

Aug 2022 – Nov 2025

Khosrow Pasha Platform for Digital Education / MindUp Academy

- Taught robotics, Arduino programming, and engineering fundamentals.
- Developed curricula, training materials, and practical learning activities.
- Delivered technical education remotely and communicated engineering concepts to students with different technical backgrounds.

3D Design & Printing Trainee

Jul 2020 – Jan 2021

SOFBIT Company · Gaziantep, Türkiye

- Contributed to feasibility research for personalized prosthetic solutions.
- Assisted with engineering design and 3D-printing prototyping activities.

Automation System Intern

Aug 2019 – Sep 2019

Merkon Otomasyon · Adana, Türkiye

- Participated in industrial automation projects alongside engineers and technicians.

LANGUAGES

Arabic

Native

English

C1

Turkish

C1

ADDITIONAL EXPERIENCE

Conference Organizer

United Engineers Organization · Aug 2018 – Sep 2018

Teacher

Syrian Students Union in Düzce · Jul 2021 – Aug 2021

Children's Gathering Organizer

Hayata Destek / Support to Life · Jul 2015